Integrity Watch

Observe change.

Verify recovery. Keep control.

01 **Native s390x CPU execution**

02 **Matched runtime integrity checks**

03 **Approved, verified recovery**

An evidence-backed control loop around AI guardrails.

THE PROBLEM

Valid scores can hide a changed control.

A changed output path can change decisions while the model still returns valid numbers.

Same model.

Same request.

Changed runtime output.

One score cannot tell us whether traffic changes—or the control changes.

ACTUAL NATIVE REPLAY EXAMPLE

Approved baseline	After ×0.25 fault
0.7426	**0.1856**

Fixed decision threshold: 0.2083

Recorded scale / seed 0 · Request index 135

Evidence: FaultBackend.score · Hash-verified scale-0 event trace · Example shows a decision change, not a new safety label

THE SOLUTION

Observe. Verify. Respond.

Separate the request decision, runtime evidence, and response authority.

01

Guardrail + canary

The guardrail gates the request.

The canary supplies a second score view.

REQUEST DECISION

02

Matched probes

Check both score vectors against a fixed, approved baseline.

RUNTIME EVIDENCE

03

Verified recovery

Execute the candidate. Verify it.

Apply only with approval.

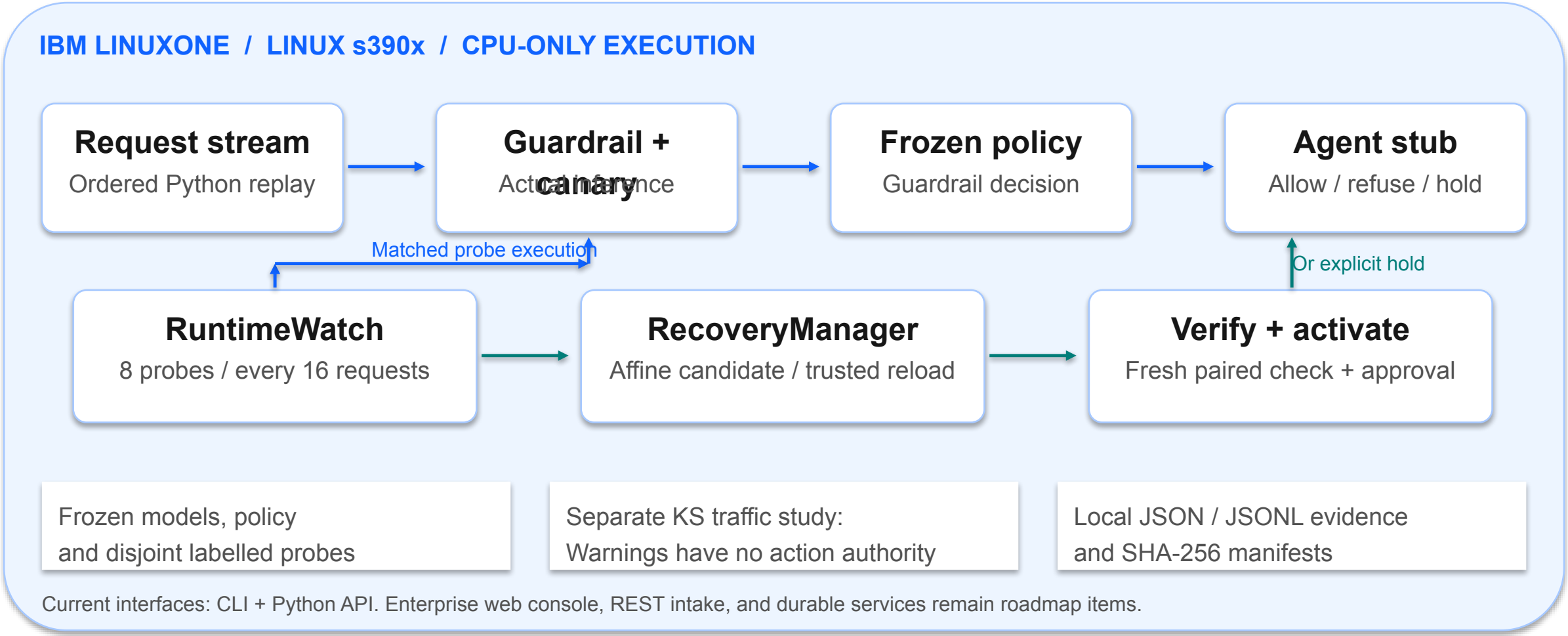
RESPONSE AUTHORITY

A traffic warning is evidence—not authorization to replace the model.

Evidence: SupervisedRuntime · RuntimeWatch · RecoveryManager · ProfiledTrafficWatch

IMPLEMENTED ARCHITECTURE

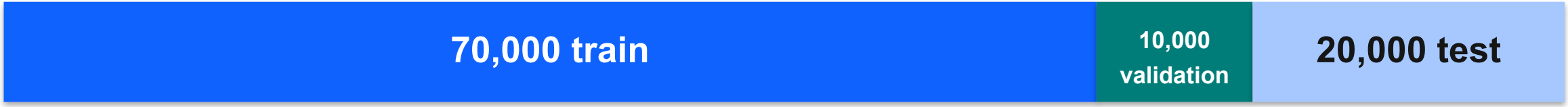
One native execution boundary.



Evidence: src/monitor/runtime.py · src/response/supervised.py · src/response/recovery.py · Full Mermaid diagram in the content kit

Controlled data. Lightweight models.

100,000 synthetic requests · 100 template families · 91 reviewed semantic groups



Related cases stay together. Only text enters the classifiers.

Guardrail

Word + character TF-IDF
Logistic regression · C=4
Frozen threshold · 0.2083

Canary

Character TF-IDF
Logistic regression · C=1
Frozen threshold · 0.2771

Healthy test · 20k

88.67% accuracy
99.96% harmful recall
22.63% safe FPR

External regression: XSTest (300) + human ToxicChat (2,853).

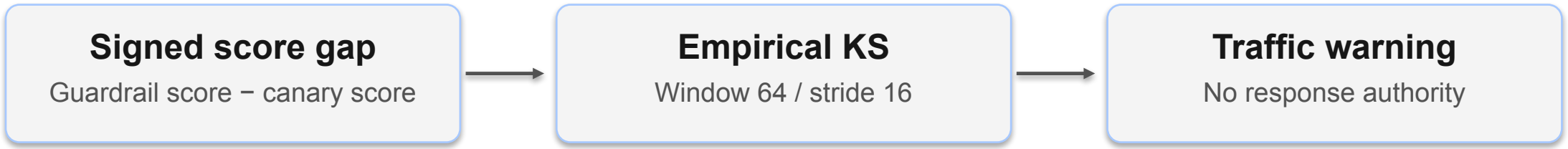
Earlier sources: WAInjectBench, JBB + mirror, StrongREJECT, HarmBench, Dolly. OASST1: acquisition only. Full roles and licenses in notes.

Evidence: data-manifest.json · model-cards.json · benchmark-report.json · Variants are not independent human requests

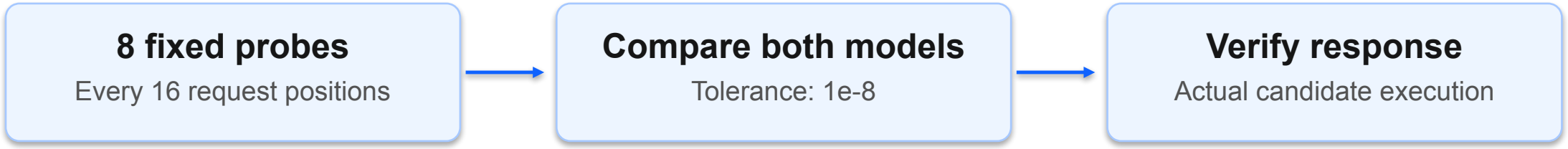
Two signals. Different authority.

Traffic change, runtime change, and labelled safety performance remain separate.

HISTORICAL TRAFFIC-STUDY PATH



SELECTED RELEASE: RUNTIME PATH



Project contribution: verified recovery, explicit holds, repeated incidents, and native reproducibility.

Cheap Canaries is prior art. Scan martingales and MMD are reference methods, not implemented in this release.

Evidence: src/monitor/disagreement.py + runtime.py · Cheap Canaries source snapshot 90726d9 · No head-to-head performance claim

PRODUCT DEMONSTRATION

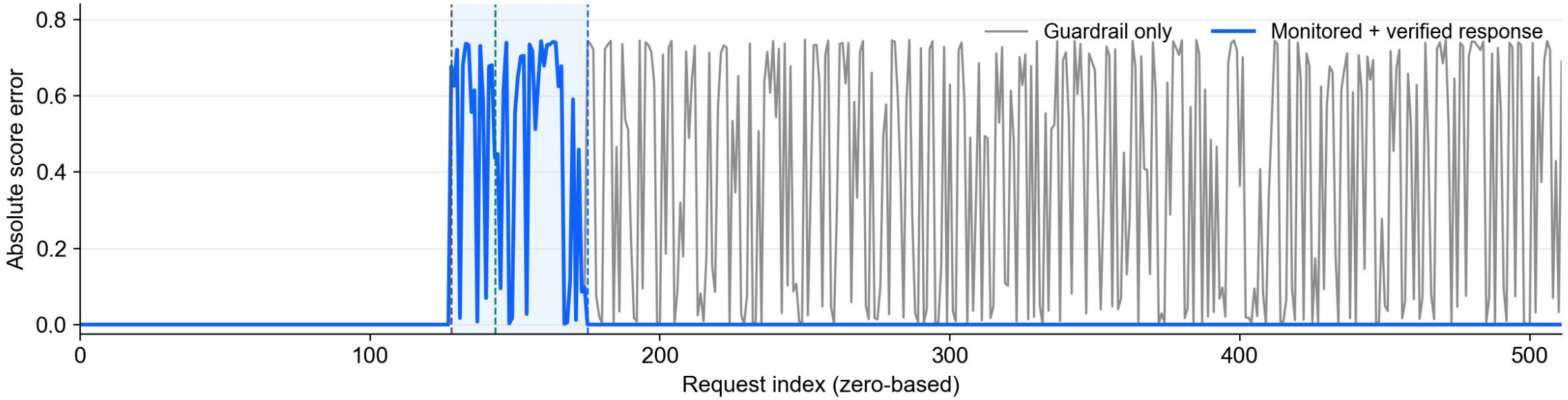
From fault to verified recovery.

Actual native scale / seed-0 trace—not a product dashboard or a simulated screenshot.

128 FAULT STARTS

143 RUNTIME CONFIRMS

175 VERIFIED CORRECTION



1,000-request endpoint: guardrail error 0.0 · canary error 0.0 · decision mismatches 0

Evidence: hash-verified scale-0 events and endpoint · 15-request confirmation delay + 32-position response delay

MEASURED RESULTS

Measure every request—not only the endpoint.

60 paired episodes · 30,720 observations · Controlled synthetic fault suite

CORRECT AUTOMATIC DECISIONS

81.69%

75.64% guardrail only → +6.05 pp

HARMFUL ALLOWS

−87.44%

4,738 → 595 / 15,840 harmful requests

Full accounting	Guardrail only	Monitored
Safe refusal rate	18.44%	**22.47%**
Held requests	0	**1,685 / 5.49%**
Runtime model-input evaluations	30,720	**152,150**

55/55 incidents confirmed · 45/50 fault episodes restored · 5 held · 0/10 false control confirmations

Evidence: benchmark-report.json + independent-audit.json · Holds receive zero automatic-correctness credit; not production-rate estimates

Verification is the activation gate.

The actual candidate must pass—not just its plan or cached output.

01

Execute

Run the candidate on separate verification probes.

02

Verify

Check both score vectors and decision outcomes.

03

Reverify

Check again before a generation-bound switch.

Explicit approval + single-use plan
Weights and policy thresholds stay fixed.

No verified replacement?
Keep an explicit hold.

Runtime restoration preserves baseline behavior. Semantic classifier quality still requires labelled evaluation.

Evidence: RecoveryManager.propose/apply · verify_outputs · SupervisedRuntime.release_hold · Full scope and historical recalibration in notes

Native IBM LinuxONE. End to end.

A complete local CPU workflow: generate → fit → infer → detect → verify → audit.

s390x. Big-endian Linux.
No external model API.
Fresh offline reproduction.

Qualified container:
Two-CPU quota · 4 GiB memory limit

39.82 s
Selected classifier fit

923.84 MiB
Peak child-process RSS

292
Native tests pass

100,000
Data rows reproduced

A guardrail needs a watcher. Every recovery needs evidence.

Native CPU qualification · 4 optional PyArrow tests skip · No Telum/Spyre, attestation, hardware-speedup, or production-capacity claim